

Talha Ahmed

Pipeline & Automation Engineer · Technical Artist

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SUMMARY

Pipeline and tools engineer and technical artist with **16+ years** building production software, tools, and automation for animation, VFX, and games. Deep in open-standard asset pipelines (**OpenUSD, MaterialX / OpenPBR**) and large-scale, cloud-backed automation, increasingly built with **AI / LLM tooling**. Ships clean, tested **Python** and **C++** across DCCs (**Houdini, Maya, Nuke, Unreal**); works remote and async across timezones and communicates clearly across engineering, art, and production.

TECHNICAL SKILLS

Languages	Python (primary), C/C++, C#, JavaScript, Lua, Bash, PowerShell, MEL/VEX
DCC & CG	Houdini (Solaris, HDA/OTL, hython, PDG/TOP, COPs), Maya, Nuke, Unreal Engine (Interchange), Blender, Arnold, RenderMan, Substance, Deadline
Interchange & Imaging	OpenUSD, MaterialX, OpenPBR, USDZ; OpenImageIO; color management, deep compositing, stereoscopy
Cloud & Services	AWS (S3, EC2, Step Functions, Deadline), FastAPI/Uvicorn, REST/WebSocket
Automation & Tooling	CI/CD, pytest, ruff, uv, Rez, Qt/PySide2, Perforce, Git, ShotGrid/ftrack, Sphinx
Data & Web	SQL, MongoDB, GraphDB, ORMs; Django, Next.js, Selenium
ML / AI	LLMs & agents (Claude Code), Stable Diffusion, TensorFlow, PyTorch, Hugging Face

EXPERIENCE

Epic Games — Quixel Contract via PulsarBoot Algorithms · Remote · Apr 2025 – Jul 2026
Pipeline & Automation Engineer — core developer & architect

Core engineer and architect of a production 3D-asset conversion and automation system built on open interchange standards, working across a cross-functional team of management, engineers, and artists.

- Built a **dual-backend** conversion pipeline on **OpenUSD** and **MaterialX / OpenPBR** — a **Houdini** engine (Solaris, custom HDAs, hython) driven from a clean Python API, and a **pure-Python** backend that reproduces its output with no DCC dependency.
- Kept the two backends in parity via an independent, automated **QA gate** (geometry / material / USD / texture validators plus optional DCC render smoke-tests) — QA-as-arbiter rather than brittle image diffing.
- Designed a **provider-based asset model** with schema validation and a metadata auto-repair registry; scaled **corpus-scale** batch tooling to tens of thousands of legacy assets with corruption detection and an auditable rejection pipeline, hardened with idempotent relaunch and a 404-aware multi-pass gather (**78.7% → 98.9%** coverage recovery).
- Wrote an in-house **FBX→USD reader** (Autodesk FBX SDK, two-pass local-space reconstruction) preserving per-node pivots; centralized color-space and OIIO-measured displacement logic shared by both backends.
- Collaborated with management, pipeline engineers, and senior artists, and ran a structured defect→rule review loop with the receiving pipeline team; remote and async across US/UK/EU/Asia. Adopted **Claude Code / LLM agents** for development and documentation.

Blink Industries London · Remote · Aug 2024 – Oct 2024
Pipeline Developer (Contract)

- Development in a hybrid 2D/3D Maya / Toon Boom Harmony / Adobe pipeline: TD tools in Maya, edit generation for Adobe Premiere, 2D sequence layout in After Effects, and 3D-object integration into Harmony.

Untold Studios London · Remote · Oct 2023 – May 2024
Pipeline Developer (Contract)

- Contributed to a VFX-heavy Maya / Houdini / Nuke pipeline; improved artist UX in data ingestion and review publishing; ran requirements analysis and feature negotiation with animation and match-move craft heads.

One Animation / Moonbug Singapore · Remote · Sep 2022 – Sep 2023
Sr Pipeline TD (Contract)

- R&D on a lean Maya / Houdini animation pipeline: auto-documentation with Sphinx, character costume and animation libraries, asset ingestion and archiving, and core pipeline configuration management with Rez.

Passion Pictures

London · Remote · Mar 2022 – Sep 2022

Pipeline Developer (Contract)

- Software-agnostic hybrid VFX/animation pipeline: rendering in Arnold, deep-compositing workflows (Arnold + Nuke), and integration of pipeline tools with ShotGrid via its Python API.

Nebula Pipeline

Karachi · May 2019 – Jan 2021

Studio Developer / Consultant (dba Nebula Pipeline)

- Consulted with animation studios on CGI production pipelines and built the **Nebula Pipeline Toolbox** — a departmentalized Maya / Redshift production pipeline spanning production planning, asset development, layout/animation, lighting/rendering, and post, with publicly published documentation.
- Authored a suite of production tools (Episode/Sequence Planner, Assets Explorer, Shader Transfer, Proxy Cache Switch, Multishot Export, Redshift AOV Tools, Scene Bundle, and more). Docs: [nebula-pipeline-toolbox-docs.readthedocs.io](https://docs.readthedocs.io).

ICE Animations

Karachi · Jan 2011 – Jan 2019

Pipeline Supervisor / CG Researcher

- Trained and led an agile team building a Maya/Nuke pipeline for feature, TV, TVC, and educational production; heavy tool development and production/infrastructure troubleshooting.
- Research: tools and plugins in Python and C++ — a RenderMan/RSR hair shader, a Tessendorf ocean-surface Maya plugin in C/C++, and dynamic-curve spider-web simulation.
- Selected screen credit: *Mansour* (animated TV series, 2012–).

Post Amazers

Karachi · Mar 2004 – Jun 2008

Technical Artist / Programmer

- Tool development, procedural animation, effects programming, and stereoscopy in RenderMan, MEL, and C/C++; built a particle-based crowd system after Craig Reynolds' work on autonomous agents.
- Selected screen credit: *Son of the Mask* (feature, 2005).

ADDITIONAL EXPERIENCE

Ranai.pk

Karachi · 2021 – 2023

Operations & Digital (family clothing business)

- Product photography, catalog production, bookkeeping, social media, and website management for a clothing-manufacturing store (business now on hold).

EDUCATION

B.S. Computer Systems Engineering — GIK Institute, Pakistan

1999 – 2003

EM CIMET (Erasmus Mundus) coursework — U. of Granada (Spain) & Gjøvik UC (Norway): color imaging & multimedia systems

2008 – 2010

P.G.D. Business Administration — IBA Karachi

2005 – 2007

CERTIFICATIONS

- **Deep Learning Specialization** — DeepLearning.ai / Coursera (neural networks, CNNs, RNNs), 2019
- **AI & GPT Applications** — Encode.club (agents, local AI deployment, vision & image generation), 2024

LANGUAGES

English (fluent) · Urdu (native) · Spanish (basic)